

Chapter 6

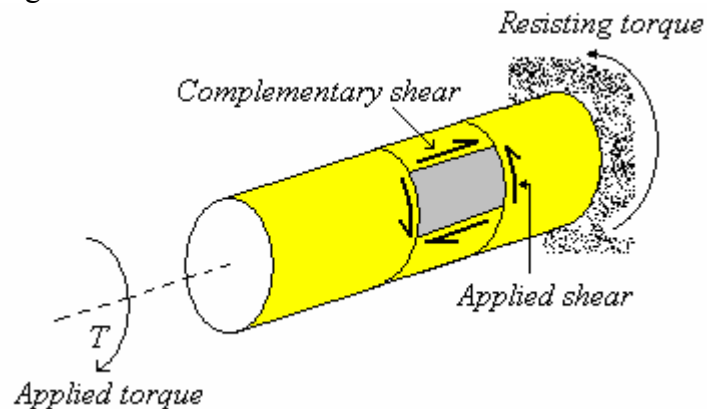
TORSION of SHAFTS

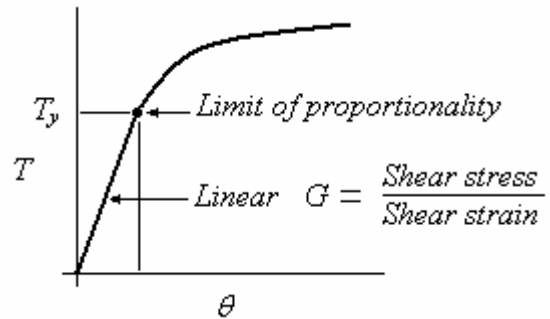
Simple Torsion Theory

When a uniform circular shaft is subjected to a torque it can be shown that every section of the shaft is subjected to a state of pure shear.

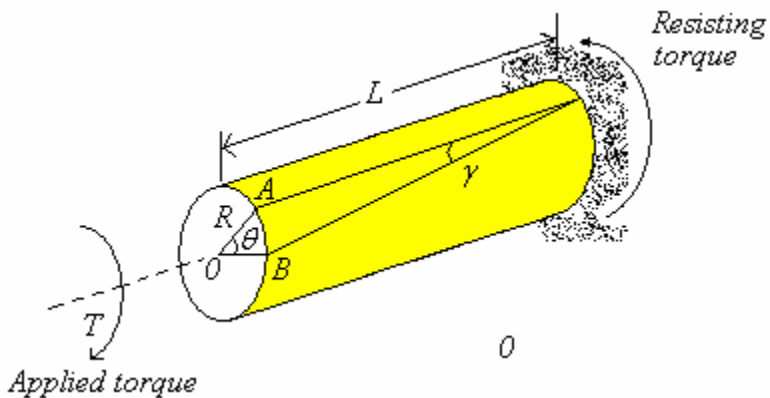
Assumptions

1. The material is homogeneous, i.e. of uniform elastic properties throughout.
2. The material is elastic, following Hook's law with shear stress proportional to shear strain.
3. The stress does not exceed the elastic limit of proportionality.
4. Circular sections remain circular.
5. Cross sections remain plane.
6. Cross sections rotate as if rigid.





Angle of Twist (θ)



After applying the torque point A moves to point B and made an angle θ .
 AB subtends an angle γ at the fixed end, γ is the shear strain.

$$\therefore \overline{AB} = R\theta = L\gamma$$

$$\gamma = \frac{R\theta}{L}$$

$$\text{Modulus of rigidity } G = \frac{\text{Shear Stress}}{\text{Shear Strain}} = \frac{\tau}{\gamma}$$

$$\therefore \gamma = \frac{\tau}{G}$$

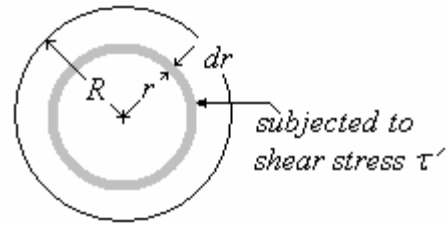
$$\therefore \frac{R\theta}{L} = \frac{\tau}{G} \Rightarrow \frac{\tau}{R} = \frac{G\theta}{L}$$

$$\frac{\tau'}{r} = \frac{G\theta}{L}, \tau' \text{ is the shear stress at any } r$$

Modulus of Rigidity (G)

It is a measure of how much deformation the material would undergo subjected to a shear. G is the property which relates to the stiffness of the material, units N/m^2 .

Stresses



The force set up on each surface = Stress $\times$ Area resisting shear

$$F = \tau' \times 2\pi r dr$$

This force will produce a moment about the centre axis of the shaft

$$T_i = \tau'(2\pi r dr)r$$

$$T_i = 2\pi \tau' r^2 dr$$

Σ All these moment is the total torque.

$$T = \int_0^R 2\pi \tau' r^2 dr$$

$$\tau' = \frac{G\theta}{L} r \Rightarrow T = \int_0^R 2\pi \frac{G\theta}{L} r^3 dr$$

$$T = \frac{G\theta}{L} \int_0^R 2\pi r^3 dr$$

The integral $\int_0^R 2\pi r^3 dr$ is called the polar second moment of area J .

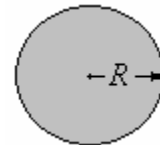
$$\therefore T = \frac{G\theta}{L} J$$

$$\frac{T}{J} = \frac{G\theta}{L} = \frac{\tau}{R}$$

- For circular solid shaft

$$J = \int_0^R 2\pi r^3 dr = 2\pi \left[\frac{r^4}{4} \right]_0^R$$

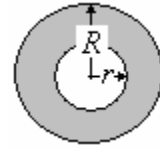
$$J = \frac{\pi R^4}{2} = \frac{\pi D^4}{32}$$



- For hollow shaft

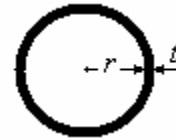
$$J = \int_0^R 2\pi r^3 dr = 2\pi \left[\frac{r^4}{4} \right]_0^R$$

$$J = \frac{\pi(R^4 - r^4)}{2} = \frac{\pi(D^4 - d^4)}{32}$$



- For thin-walled hollow shafts

$$J = 2\pi r^3 t$$



Section Modulus

At $r = R$, τ is maximum:

$$\tau_{\max} = \frac{T}{J} R \Rightarrow \tau_{\max} = \frac{T}{Z}$$

Where Z is polar section modulus $= \frac{J}{R} = \frac{2J}{D}$

- For solid shaft $J = \frac{\pi D^4}{32} \Rightarrow Z = \frac{2J}{D} = \frac{\pi D^3}{16}$
- For hollow shafts $Z = \frac{\pi(D^4 - d^4)}{16D}$

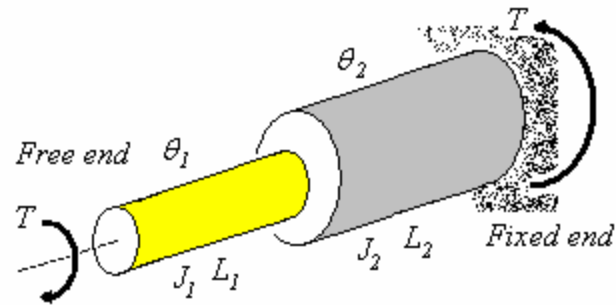
Torsional Rigidity

The angle of twist per unit length of shafts is given by:

$$\frac{\theta}{L} = \frac{T}{GJ}, GJ \text{ is termed torsional rigidity.}$$

Composite Shafts

- *Series Connection*



When two similar or dissimilar shafts of the same or of different materials are connected together to form one composite shaft, the driving torque being applied at one end and the resisting torque at the other, the shafts are said to be connected in series.

Consider each component shaft separately, applying the torsion theory to each in turn. The composite shaft will therefore be as weak as its weakest component.

$$T = T_1 = T_2$$

$$T = \frac{G_1 \theta_1 J_1}{L_1} = \frac{G_2 \theta_2 J_2}{L_2}$$

$$\theta_{Net} = \theta_1 + \theta_2$$

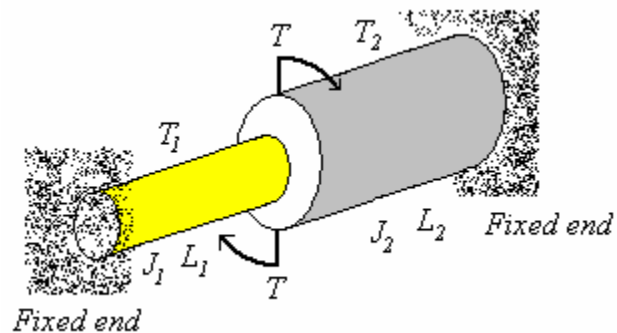
$$\theta_{Net} = \frac{TL_1}{G_1 J_1} + \frac{TL_2}{G_2 J_2}$$

If $\theta_1 = \theta_2$ same angle of twist for the same material:

$$\frac{J_1}{L_1} = \frac{J_2}{L_2}$$

$$\frac{L_1}{L_2} = \frac{J_1}{J_2}$$

- **Parallel Connection**



If the driving torque is applied at the junction of the two shafts connected together and the resisting torque at the other ends of the two shafts, then the shafts are said to be connected in parallel.

The torque applied in this case is divided between the two shafts but the angle of twist is the same for each shaft.

$$T = T_1 + T_2$$

$$\theta_1 = \theta_2$$

$$\therefore \frac{T_1 L_1}{G_1 J_1} = \frac{T_2 L_2}{G_2 J_2}$$

If $L_1 = L_2$

$$\frac{T_1}{T_2} = \frac{G_1 J_1}{G_2 J_2}$$

Maximum stress in each part:

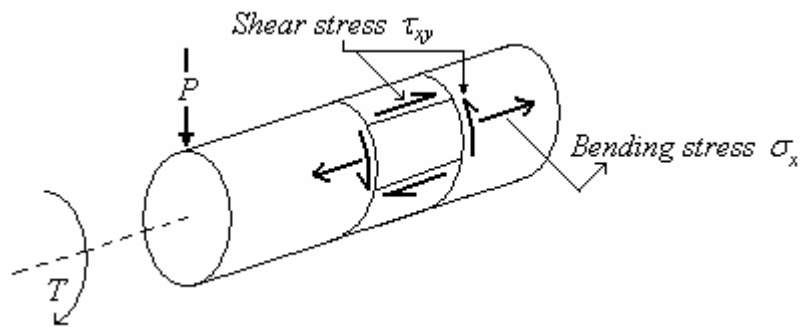
$$\tau_1 = \frac{T_1 R_1}{J_1} \text{ and } \tau_2 = \frac{T_2 R_2}{J_2}$$

Power Transmitted by Shafts

If a shaft carries a torque $T \text{ N.m}$ and rotates at $w \text{ rad/s}$ it will do work at the rate:

$$P = Tw \text{ watts}$$

Combined Bending and Torsion



Shafts carry torque and subjected to bending.

To find the principal stresses:

$$\sigma_1 \text{ or } \sigma_2 = \frac{1}{2} \left\{ (\sigma_x + \sigma_y) \pm \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau_{xy}^2} \right\}$$

The shaft can be considered to be subjected to equivalent torques or equivalent bending moments.

$$\tau_{\max} = \frac{1}{2} (\sigma_1 - \sigma_2) = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2} \right)^2 + \tau_{xy}^2}$$